



Assignment - 1

Course : CSE250

Student name: MD Rakib Hossain Ontu

Student Id : 22101879

Theory section : 16

G-suit : rakib.hossain.ontu@g.bracu
.ac.bd

Course Instructor: Ramkrishna Saha

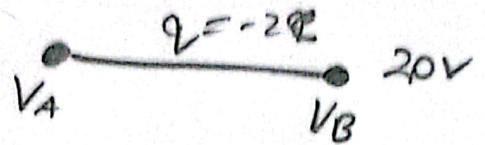
Ans to ques - 01

Given, $W = 10\text{ J}$

$$q = -2\text{ C}$$

$$V_B = 20\text{ V}$$

$$V_A = ?$$



So, $W = q(V_B - V_A)$

$$\frac{W}{q} = V_B - V_A$$

$$\Rightarrow V_A = -\frac{W}{q} + V_B$$

$$= -\frac{10}{-2} + 20$$

$$= 25\text{ V} \quad \underline{\text{(ans)}}$$

Ans to ques - 2

Given, $q = 2 - Ae^{-\frac{At}{2}}$

We know,

$$I = \frac{dq}{dt} = \left(-\frac{A}{2}\right)e^{-\frac{At}{2}}$$

at, $t = 2$, and $A = 79$,

$$I = \left(-\frac{79}{2}\right)e^{-\frac{79 \times 2}{2}}$$

$$= 3013.12\text{ mA}$$

Ans to ques - 3

In the given circuit,

$$P_1 = 6 \times (-30) = -180 \text{ W}$$

$$P_2 = 6 \times (12) = 72 \text{ W}$$

$$P_3 = 2 \times (28) = 56 \text{ W}$$

$$P_4 = 1 \times (28) = 28 \text{ W}$$

$$P_5 = 3 \times 18 = 54 \text{ W}$$

$$P_6 = 3 \times (-5 \times 1) = -30 \text{ W}$$

Here, only P_1 and P_6 are suppliers and other are absorbers.

Ans to ques - 04

Here, $R_2 - R_5$ all are in parallel.

$$\begin{aligned} R_p &= \left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5} \right)^{-1} \\ &= \left(\frac{1}{20} + \frac{1}{10} + \frac{1}{5} + \frac{1}{40} \right)^{-1} \\ &= \left(\frac{2+4+8+1}{40} \right)^{-1} \\ &= \left(\frac{15}{40} \right)^{-1} \\ &= \frac{40}{15} \text{ k}\Omega \end{aligned}$$

We know, for the given circuit,

$$I_o = I_x \times \frac{\frac{1}{R_o}}{\frac{1}{R_p}}$$

$$\Rightarrow I_x = \frac{I_o R_o}{R_p} \quad [I_o = 79A]$$

$$= \frac{79 \times (40)}{\frac{40}{15}}$$

$$= 1185A$$

Ans to ques no - 5

After filling the data in the circuit,

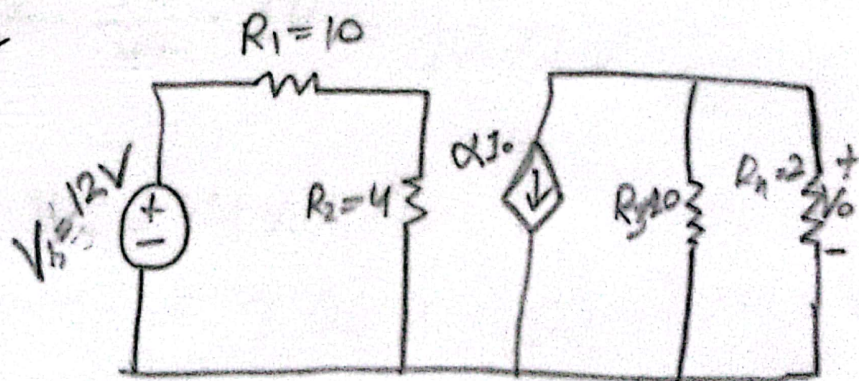
$$R_1 = R_3 = (1+9) = 10k\Omega$$

$$R_2 = (2+2) = 4k\Omega$$

$$R_4 = 2k\Omega$$

$$V_s = (3+9) = 12V$$

$$\alpha = (2+2) = 4$$



R_1 and R_2 are in series, so, $R_{\text{series}} = (10 + 4)$
 $= 14 \text{ k}\Omega$

R_3 and R_4 are in parallel so, $R_p = \left(\frac{1}{R_3} + \frac{1}{R_4} \right)^{-1}$
 $= \left(\frac{1}{10} + \frac{1}{2} \right)^{-1}$
 $= \left(\frac{1+5}{10} \right)^{-1}$
 $= \left(\frac{6}{10} \right)^{-1}$
 $= \frac{5}{3} \text{ k}\Omega$

So, Total resistance, $R_{\text{eq}} = R_s + R_p$
 $= 14 + \frac{5}{3}$
 $= \frac{47}{3} \text{ k}\Omega$

So, $I_o = \frac{V_o}{R_{\text{eq}}} = \frac{12}{\frac{47}{3}} = \frac{36}{47} \text{ A} = 0.76 \text{ A}$ (ans)

So, $V_o = I_o \times R_{\text{series}} = \frac{36}{47} \times 14 = 10.72 \text{ V}$ (ans)

Voltage across R_2 , $V_{R_2} = 4 \times 0.76$
 $= 3.06 \text{ V}$ (ans)

Ans to ques - 06

Applying KBL in loops,

~~Loop 1~~,

$$3 + V_1 - V_3 = 0 \quad \text{--- (i)}$$

$$V_1 - V_2 - 2 = 0 \quad \text{--- (ii)}$$

$$-4 + V_3 + V_4 = 0 \quad \text{--- (iii)}$$

$$V_4 + 2 + 5 = 0 \quad \text{--- (iv)}$$

from (iv), $V_4 = -7 \text{ v}$

from (iii), $-4 + V_3 - 7 = 0$
 $\Rightarrow V_3 = 11 \text{ v}$

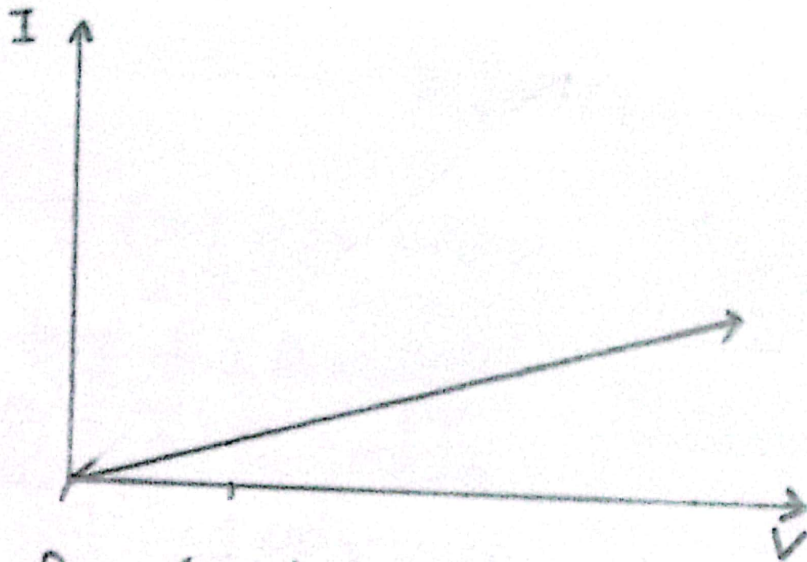
from (i), $3 + V_1 - 11 = 0$
 $\therefore V_1 = 8 \text{ v}$

from (ii), $8 - V_2 - 2 = 0$
 $\Rightarrow V_2 = 6 \text{ v}$

So, Voltage across the branches,

$$V_1 = 8 \text{ v}, V_2 = 6 \text{ v}, V_3 = 11, V_4 = -7$$

Ans to ques - 07



$$\text{for } B = (7+9) = 16$$

$$V = IR = 16I$$

Ans to ques - 08

$$\text{Here, } I = 5A$$

$$V_x = 25V$$

$$R_{vx} = ?$$

$$\text{So, } 5R + R(-25) - 25 = 0$$

$$\Rightarrow R = \frac{25 - 5R}{-25}$$

Ans to ques - 09

Total resistance, $R_{eq} = (6 + 3 + R) = 9 + R \Omega$

$$\text{So, } I(9 + R) = 140 \text{ V}$$

$$\therefore 9I + IR = 140$$

$$\text{Or, } I = \frac{140}{9 + R}$$

Now, for, 100 V difference,

$$I(R + 6) = 100$$

~~$$\Rightarrow 6I + IR = 100$$~~

$$\Rightarrow \frac{140}{9 + R} (6 + R) = 100$$

$$\Rightarrow 840 + 140R = 900 + 100R$$

$$\Rightarrow 40R = 60$$

$$\Rightarrow R = 1.5 \Omega$$